DUSI VENKATA SAI KARTHIKEYA

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Education

Andhra University College of Engineering *Visakhapatnam, India*
B.Tech in Information Technology | **CGPA:** 8.4/10 *2023 – 2027*
Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems, Operating Systems, Machine Learning

Experience

Software Engineering Intern — My TC *Sep 2026 – Present*

- Studied the **real-estate transaction coordination domain** and analyzed the Listing TC / seller-side workflow from executed contract through closing.
- Analyzed the **escrow opening workflow**, including the transaction information required to initiate escrow with the preferred escrow company.
- Analyzed the **seller disclosure workflow**, including document completion, review, delivery to the buyer side, and follow-up for pending signed disclosures.

Skills

Languages: Python, JavaScript, SQL
Web Development: HTML, React, Tailwind CSS, Node.js, FastAPI, REST APIs
Databases & Authentication: MySQL, Firebase, OAuth, Clerk
Computer Vision: OpenCV, YOLO (v5/v8), Ultralytics, ONNX Runtime
Tools & Platforms: Git, GitHub, Postman, Figma, Docker, AWS, Vercel, Render, Claude Code

Projects

Attendly — Offline-First Attendance Management PWA *Live Demo — GitHub*

- Built an offline-first attendance tracking PWA using Next.js and TypeScript, with localStorage as the primary data store and optional Firebase sync; used in production by **20 students** across **3 academic semesters**.
- Designed the core attendance-calculation logic, weighting classes by credit hours to drive analytics, bunk-risk indicators, recovery-timeline estimates, and semester management, with business logic kept separate from the UI.
- Replaced a cloud-based AI timetable import with client-side OCR (Tesseract.js) and deterministic parsing, then maintained the app through real student usage, resolving persistence, rescheduling, and sync issues reported by users.

Technologies: React, Next.js, TypeScript, Firebase, Vitest, Tailwind CSS, Vercel

AegisLogix — Industrial Computer Vision System *Live Demo — GitHub*

- Built a full-stack container damage inspection system: FastAPI backend with REST APIs and a React frontend, handling image upload validation, preprocessing, and interactive damage visualization.
- Fine-tuned a YOLOv5s model for damage detection, served via an ONNX Runtime inference pipeline; containerized the backend with Docker and deployed on Render.
- Improved overall **mAP50 from 0.217 to 0.561** through transfer learning, dataset engineering, and iterative fine-tuning; raised the **Dent class mAP50 to 0.83** by fixing incorrect annotations and addressing class imbalance.

Technologies: Python, FastAPI, React, YOLOv5, OpenCV, ONNX Runtime, Tailwind CSS, Render

GitRunByKaru — Open-Source npm CLI Package *Live Demo — npm — GitHub*

- Built and published an open-source npm CLI package (**1,500+ downloads**) that clones a repository, detects its stack, installs dependencies, and launches the project with a single command.
- Designed a CLI runtime that automates framework detection, verifies service readiness over HTTP, and manages temporary workspace lifecycles, with MCP integration so AI coding agents can invoke the tool directly.

Technologies: Node.js, JavaScript, npm, Git, MCP

Achievements

DP World Hackathon 2026 *April 2026*

- Qualified for **Round 2** of DP World Hackathon 2026, hosted at BITS Pilani Hyderabad.

MumbaiHacks 2025 *Nov 2025*

- Qualified for **Round 3**, finishing among the top 150 teams out of **3,000+ participants**.